

ASSIGNMENT 1 REPORT - RELIABLE UDP COMMUNICATION IN MININET

1. STUDENT DETAILS

- * Name: Sonkalpa Borah
- * Roll No: CS-BTC24-08
- * Timeout used (roll last digit 8): 1.5 seconds

2. OBJECTIVE

The objective of this assignment is to implement reliable message transfer over UDP using stop-and-wait logic. Reliability is added by attaching sequence numbers, waiting for ACK packets, applying timeout, and retransmitting packets when ACK is not received.

3. TOPOLOGY AND SETUP

- * Topology: 'h1 -- s1 -- h2'
- * 'h1' runs the UDP server and 'h2' runs the UDP client.
- * Client message format: 'SEQ|MESSAGE'
- * Server ACK format: 'ACK|SEQ'

In this run environment (WSL), packet-loss behavior was executed using the provided application-level fallback mode ('--emulate-loss') because kernel 'tc/netem' was unavailable.

4. PROGRAM DESIGN

CLIENT

- * Sends one message at a time.
- * Waits for matching ACK for current sequence number.
- * If timeout occurs or ACK is lost/invalid, retransmits the same sequence.
- * Tracks total sent packets, retransmissions, and total transfer time.

SERVER

- * Receives 'SEQ|MESSAGE' packets.
- * Sends 'ACK|SEQ' for every received packet.
- * Maintains set of seen sequence numbers.
- * Counts unique packets and duplicates.

5. RESULTS

From 'result_table.csv':

```
loss_percent | timeout | total_messages | total_packets_sent | total_retransmissions |  
transfer_time_seconds | status  
--- | ---: | ---: | ---: | ---: | ---  
0 | 1.5 | 10 | 10 | 0 | 0.304568 | SUCCESS  
5 | 1.5 | 10 | 10 | 0 | 0.305424 | SUCCESS  
10 | 1.5 | 10 | 11 | 1 | 0.335629 | SUCCESS
```

Observation: as loss increases to 10%, one retransmission appears and transfer time increases.

6. SERVER-SIDE DUPLICATE OBSERVATION

From 'logs/server_loss10.out.txt', the server detected one duplicate packet ('seq=4') and still completed with:

```
* 'TOTAL_UNIQUE_MESSAGES_RECEIVED=10'  
* 'TOTAL_DUPLICATES_DETECTED=1'  
* 'STATUS=SUCCESS'
```

7. REQUIRED ANSWERS

1. Why is ACK needed?

- ACK confirms successful delivery of a specific sequence number, so the sender knows when it can move to the next packet.

2. Why is timeout needed?

- Timeout allows the sender to recover when data or ACK is lost by triggering retransmission after waiting for a bounded period.

3. Why can duplicate messages occur?

- If ACK is lost, the sender retransmits the same packet. The receiver may already have processed it, so the same sequence number arrives again.

4. What happens when packet loss increases?

- Retransmissions increase, transfer time increases, and protocol overhead increases, reducing effective performance.

5. How is this method similar to TCP?

- It uses reliability concepts similar to TCP: sequence tracking, acknowledgment, timeout, and retransmission.

6. How is it different from TCP?

- This implementation is minimal stop-and-wait logic without TCP features like sliding window, congestion control, flow control, and byte-stream management.

8. SCREENSHOTS

Add these screenshots in 'screenshots/' and include them in the final PDF:

- * 'nodes.png'
- * 'net.png'
- * 'pingall.png'
- * 'server_output.png'
- * 'client_output.png'

9. CONCLUSION

The reliable UDP implementation correctly delivered all 10 messages across tested loss profiles, detected duplicates at higher loss, and showed expected increase in retransmission/time with loss.